



Inspiring Excellence

CSE423: Computer Graphics

Project Title: Zombie Combat

Section: 11

Group: 5

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Zombie Combat

In this immersive closed-world survival experience, the players are thrust into a high-stakes environment where every resource counts and every shadow hides a threat. As the Hero, your primary goal is simple: survive. Caught within a confined territory teeming with diverse undead mutations and physical hazards, you must master the art of scavenging, tactical combat, and environmental manipulation.

Key Game Features:

1. Tactical Projectile Combat

The **Left Mouse Button** throws knives affected by simulated gravity, causing them to travel in a downward arc. Knives disappear immediately upon hitting the ground or striking a zombie, dealing 1 HP of damage per hit.

2. Close Combat

The **Right Mouse Button** activates a melee strike. This functions as an area-of-effect attack, dealing 1 HP of damage to all enemies within a 180-unit radius of the hero. This requires the player to manage their spacing carefully to hit enemies without being bitten. If the player takes damage then the player's body turns red.

3. Health Acquisition System

Health is primarily restored in two ways. Consuming apples with an **E** button that randomly spawn under trees instantly restores 15 HP. Additionally, standing within the 300 unit safe zone of a crafted bonfire provides continuous health regeneration, healing the player for 2 HP every 0.5 seconds.

4. Crafting & Resource Management

Environmental interaction yields random survival resources. The player can chop with the **X** button. Chopping trees has a 70% chance to drop usable timber, while trees also have a 70% chance to generate apples underneath them.

5. Two Game Areas and Solid Obstacles

The game world is split into two different places: a Forest (with trees, bushes, and logs) and a City (with buildings and barricades). Players and zombies cannot walk through walls, buildings, trees, or the edges of the map. The player needs to jump over the obstacle.

6. Dynamic Traversal

The player can easily move around the map and jump over things that are in the way. When the player jumps (by the click of the SPACE button), it looks and feels natural. The player moves up and forward at the same time and comes back down due to gravity. This lets them smoothly jump right over city barricades and forest logs.

7. Star Power-Ups for Full Health

Floating stars appear in the game for a short time to help the player heal when they are hurt. Stars randomly pop up near the player. If the player runs over to the star and eats it before the timer runs out, their health goes all the way back up to 100%.

8. Zombies in Bonfire & Collect Wood (from Timber)

The player can interact with the environment to collect items and instantly kill enemies. If the player gets close enough to a fallen log, they can chop it for a chance to collect wood. Also, if the player builds a bonfire, any zombies that are standing too close to that spot will burn up and die right away.

9. Atmospheric Weather Control

Unlike standard survival games, the player has the unique ability to shift the game's weather with the "P". This can be used to alter the mood of the world or potentially influence visibility during combat. The weather is rainy, snowy and sunny.

10. Diverse Enemy Hierarchy

The world is populated by two types of zombie types, each featuring unique visual designs and varying health pools. Players must identify the threat level of each creature instantly and decide whether to engage or retreat based on their current arsenal.

11.The Bonfire Sanctuary

Players can craft and place **Bonfires**, which act as temporary safe zones. These regions create a protective aura that enemies cannot enter, while simultaneously providing the Hero with vital health regeneration.

12. Scavenging & Random Loot Drops

Killing enemies will generate Dead Meat, which will increase health by acquiring it. The score is increased per Dead Meat consumption.

13. Advanced Hero Animations

Every strike or throw is visually represented, hitting or throwing knives create hand animation of the hero. At the same time, leg movement animation along with displacement.